Prevalence of anaemia and iron deficiency in three trimesters in rural Vellore district, south India

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TROPICAL DOCTOR, 2001, **31**, 86–89

SUMMARY A multistage sampling technique was used to select 845 pregnant women from two blocks in Vellore district. Haematological measurement of haemoglobin (Hb) was done on all women and serum ferritin (SF) on a subsample of 445. The prevalence of anaemia (Hb <11 g/dl) was 56.6%, 70.2% and 69.5%, respectively among the first, second and third trimester women. The mean Hb of 10.7 g/dl was significantly higher among the first trimester than among the second and third trimester women, which was less than the recommended value of 11 g/dl. Iron deficiency (SF <12 μg/L) was significantly ($P<0.05$) more among the third trimester women than among the first. The high prevalence of anaemia in each trimester in pregnancy indicates the need for iron supplementation as early as possible starting from the fourth month of pregnancy.

Introduction

Anaemia in pregnancy has caught the attention of policy planners for many decades[1]. Many studies of anaemia in pregnancy measure mean haemoglobin (Hb) concentration as an average during the entire period of pregnancy. It is well known that there are changes in the concentration during successive trimesters with the lowest at the start of the second trimester. During the first trimester of pregnancy iron requirements are lower due to the cessation of menses, and iron stores may even increase[2]. At around 16 weeks of gestation maternal blood volume and red cell mass expand so that iron requirements increase substantially[3] and the need for iron increases almost linearly until term. Even with iron supplementation Hb concentration decreases as the gestational age increases after 14 weeks and then increases to the same level as that in the first trimester[4]. Because of haemodilution during the second trimester, the cut off values (g/dl) for anaemia for pregnant women in trimester 1, 2 and 3, respectively, are 11.0 g/dl, 10.5 g/dl and 11.0 g/dl[5]. Serum ferritin levels fall dramatically below 20 μg/L during the second and the third trimester, even in women taking iron supplementation[6].

Most studies have been conducted in hospital settings, and those carried out in the community have usually been prevalence studies or have measured mean haemoglobin concentration. Few community-based studies on anaemia and Hb levels during successive trimesters have been published. Ours was a pre-intervention study as part of a community trial to prevent anaemia in pregnancy in a rural community of Tamil Nadu, India.

Methods and materials

As part of a multi centric study, a total of 895 pregnant women were randomly selected from two blocks, of which 464 women were from K V Kuppam block (KVK) and 431 were from Gudiyatham block (GB), in Vellore district during June 1996. A multistage sampling technique was used to select the required sample of pregnant women. Fifty per cent of the panchayats from each block was selected using simple random technique in the first stage and all the pregnant women were the subjects for the Hb measurements. Fifty per cent of the blood samples were selected using systematic sampling and sent for serum ferritin (SF) test. The socio economic status determined by the type of roof, the number of rooms they have, ownership of land, family size, education of the women and husband were similar having no statistical difference between the blocks. Baseline data was pooled together for analysis. Those who did not have information on gestational age (50) were omitted from the analysis. Thus, 845 pregnant women had their haematological status assessed using Hb concentration and 445 SF concentration. The Hb concentration was assessed by measuring anticoagulated venous blood and was estimated by the cyanmethemoglobin method[7]. Hb was tested in the RUHSA laboratory with 5% validation tests done at Christian Medical College and Hospital (CMC&H). Serum ferritin measurement was conducted at the Clinical Biochemistry Department of CMC&H each day. The samples were stored and tested in batches by using the coat-a-count IRMA immunoradioactive assay. Quality control was built into the test.

Results

Of the respondents 49.3% were living in houses with thatched roof, 27.9% with tiled roof and 22.7% with terraced roof. Among them 41.3% had only one room, 37.8% had two rooms and only 20.9% had three rooms in their houses. About half of them (52.7%) had less than Rs 500 as their monthly income, 36.9% had Rs 501 to 1000 and 10.2% had more than Rs 1000. A quarter (26.5%) of the pregnant women were aged 14–19 years, almost half (47.1%) in 20–24 years and another quarter (26.4%) were above the age of 25 years. Over a quarter of the respondents (26%) had less than fifth standard education, 26% had sixth to eighth standard, 18.7% had above ninth standard and the rest (29.3%) did not go to school.

The prevalence of anaemia (Hb <11 g/dl) during each trimester was 56.1%, 73.4%, 68.8%, in KVK, 57.1%, 65.8%, 70.1% in GB and 56.6%, 70.2%, 69.5% in the pooled data of both blocks.

During the first trimester there were no women with severe anaemia. The highest proportion of severe anaemia was seen during the second trimester. In the second trimester moderate anaemia was present in 35.4%, mild anaemia in 30.3% and normal Hb concentration was 29.8%. Very severe anaemia was seen in 4.5% of women. In the third trimester severe, moderate and mild anaemia was present in 2.8%, 36.9%, 29.8% of women, respectively. Thirty one per cent of women had a normal Hb. The highest mean Hb was in the first trimester, followed by third and second trimesters, respectively (Table 1).

Table 1 Prevalence of anaemia [mean, standard deviation (SD)] and range of haemoglobin (Hb) among pregnant women

Hb level in g/dl	*I Trimester*		*II Trimester*		*III Trimester*		*Total*	
	N=76	(%)	N=376	(%)	N=393	(%)	N=845	(%)
Severe ⩽6.99	–	(–)	17	(4.5)	11	(2.8)	28	(3.3)
Moderate 7.00–9.99	19	(25.0)	133	(35.4)	145	(36.9)	297	(35.1)
Mild 10.00–10.99	24	(31.6)	114	(30.3)	117	(29.8)	255	(30.2)
Normal >11.0	33	(43.4)	112	(29.8)	120	(30.5)	265	(31.4)
Mean (SD)	10.7	(1.3)	10.1	(1.5)	10.2	(1.5)	10.2	(1.4)
95% Confidence interval	10.4–11.0		9.9–11.0		10.0–10.3		10.1–10.3	
Range	7.5–13.8		4.7–14.1		3.0–13.2		3.0–14.1	

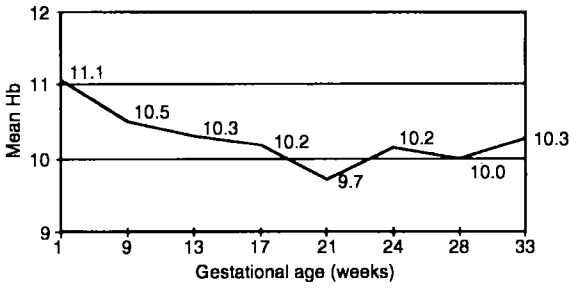

Fig. 1 Mean haemoglobin (Hb) in different gestational age, 1966

Mean Hb showed a 'V' shaped response as the gestational age increases (Figure 1).

Unlike changes in Hb concentration the SF levels continued to decrease as the trimesters increased. The prevalence of iron deficiency (SF $<12\,\mu g/L$) was greater among women in the third trimester (40.7%) than in the second trimester (35.1%), but the difference did not achieve statistical significance. However, prevalence of iron deficiency was significantly less ($P<0.05$) among those in the first trimester (18.2%) than in the third (Table 2).

The mean serum ferritin decreases as pregnancy progresses. The mean SF concentration of $36.40\,\mu g/L$ in the first trimester is again significantly ($P<0.001$) higher than those in the second and third trimesters $23.47\,\mu g/L$ and $21.99\,\mu g/L$, respectively. Three women in the second and four women in the third trimesters had SF concentration of $0.01\,\mu g/L$ which indicated absence of iron stores. The mean SF did not change significantly even if these extreme lower values were excluded (see Table 2).

The minimum Hb values among the women in the first trimester was 7.5 g/dL which was higher than those who were of higher gestational age. Minimum values of SF concentration decreases as the gestational age increases.

Discussion

Women should ideally be selected by stratified random sampling of the population of the prevalence of anaemia in different trimesters is to be reliably estimated. However, this is usually not possible and most investigators have studied the populations most accessible to them, generally hospital and clinic population.

Even the collaborative studies supported by WHO[8], to study the relative prevalence of deficiency states and anaemia in different countries, undertaken in Mexico, Poland, Israel, Venezuela, Northern India (Delhi), Southern Indian (Vellore), Burma and in seven Latin American countries suffer from sample selection among clinic population. Our study overcame this and our result reflects a true prevalence among women in different trimesters in a rural community.

In the first collaborative studies supported by WHO, women in the third trimester of pregnancy were chosen for study, since, because of their increased nutrient requirements, they are the group most at risk in the community[1]. In the above study it is documented that the prevalence of anaemia (Hb <11 g/dL) was high in India with 80% in rural North India among the women in third trimester[9] and 57.4% in mainly urban Southern India[8,9]. After three decades, with a substantial government prophylaxis programme both blocks and with additional inputs given for about two decades since 1977 in the study area of KVK which is 25 km from Vellore, the prevalence in the third trimester in 1996 was 69.7%.

Kuizon[10] observed that the Hb concentration for each trimester among Filipino pregnant women followed a Gaussian distribution. The present study also confirms this. Many studies[11–13] have shown a highly positive, significant correlation between hemoglobin concentration in early or midpregnancy and those at term, with or without the presence of iron supplementation. The prevalence of anaemia in pregnant women in their first visit to health centres of a metropolitan area, Brazil, was assessed by Guerra *et al.*[14], who documented that the prevalence of anaemia in the first trimester (3.6%) was significantly lower than in the second (20.9%) and third trimesters (32.1%). In this study although the prevalence is lower in the first trimester it is approximately the same in the second and third trimesters.

Sharma *et al.*[15] documented that all pregnant women in the age group of 13–19 years and in the second and third trimesters in a primitive tribal community of Rajasthan selected randomly from 15 villages of four blocks were

Table 2 Prevalence of iron deficiency in three trimesters

Serum ferritin	*I Trimester*		*II Trimester*		*III Trimester*		*Total*	
	N=33	(%)	N=191	(%)	N=221	(%)	N=445	(%)
$<12\,\mu g/L$	6	(18.2)	67	(35.1)	90	(40.7)	163	(36.6)
Mean	36.40		23.47		21.99		23.69	
Standard deviation	25.22		19.25		20.99		20.43	
Range	3.00–96.10		0.10–96.30		0.10–153.0		0.10–153.0	

suffering from moderate to severe anaemia. In this study the prevalence of severe anaemia is low, occurring only in the second and third trimesters with a large proportion of mild anaemia in all three trimesters. Teenagers might be iron deficient due to the demands of teenage growth spurt. Nevertheless there was no significant difference in the anaemia prevalence between rural pregnant women aged 14–19 years and in higher age groups. Conversely the mean Hb was significantly ($P<0.01$) higher than those in the 20–24 years and those aged >25 years.

A normal serum ferritin value in the first trimester does not necessarily guarantee adequate iron status in the second and third trimesters[3]. Serum ferritin, which is a reliable indicator of the iron status in the first trimester, becomes less reliable after the 20th week due to the physiological dilution of the plasma and a concurrent fall in haemoglobin and SF. For this study an SF value of $<12\,\mu g/L$ was used to define iron deficiency[16].

Ferritin concentration decreases substantially during the second and third trimesters owing to hemodilution and increased iron utilization, but the usual adult cutoff for ferritin ($<10\,\mu g/L$) is not corrected for pregnancy. Knight[17] observed through a prospective observational study that SF during the third trimester declined to 58% of the first trimester concentration among the African American women aged 16–35 years and in multiparous women.

In the mid-eighties Hercberg[18] recorded a virtual absence of iron reserves with SF $<1\,\mu g/L$ in 20%, 60% and 75% of French women during the third, sixth and ninth month of pregnancy, respectively. In our study there were no women with absence of iron reserves in the first trimester and a negligible number in the second and third trimesters.

Bentley[6] recommended that diagnosis of iron deficiency can be confirmed by SF falling to $<12\,\mu g/L$, that $<50\,\mu g/L$ in early pregnancy is an indication for iron supplements. If SF $>80\,\mu g/L$ at booking then the women are unlikely to require iron supplements. In our study 75.8% of the women in the first trimester had SF $<50\,\mu g/L$ and this could be an indication for iron supplements. At the same time, only 9.1% and 2.6% of the women in the first and second trimester had SF $>80\,\mu g/L$ who are unlikely to require iron supplements. This clearly indicates that a prophylactic iron supplementation programme should be effectively carried out in this population.

De Silva *et al.*[19] reported that depleted iron stores with SF $<12\,\mu g/L$ were found among 33.3% women of 10–26 weeks of gestation and there was no significant change even among women at >32 weeks of gestation in plantation workers in Sri Lanka. The prevalence of iron deficiency, based on a serum ferritin of $<12\,\mu g/L$ was highest at term and lowest during the first trimester indicating a decrease in iron stores as pregnancy progressed among Filipino women[20]. Our study documents 40.7% among those >28 weeks of gestation in a rural area. Perlas *et al.*[20] had recorded a mean SF value of $9.3\,\mu g/L$ [(standard deviation) SD $2.6\,\mu g/L$] and $7.1\,\mu g/L$ (SD $2.2\,\mu g/L$) during second and third trimesters which were significantly lower compared to that in the first trimester $22.6\,\mu g/L$ (SD $2.2\,\mu g/L$). In our study also the mean SF was highest in the first trimester and the lowest in the third trimester. However in all three trimesters, it was more than $20\,\mu g/L$.

WHO recommended that if the prevalence of anaemia is more than 40% that region is categorized as high magnitude of prevalence. A high magnitude of anaemia prevalence in each of the trimesters in pregnancy clearly indicates the need for prophylactic treatment with iron as soon as possible starting from the fourth month of pregnancy.

Acknowledgements

The authors would like to thank Dr V Sampathkumar, Mr Jeevan, Mr Paranthaman, Mr Venkatathiri and Mr Srinivasan for their contributions.

This publication was made possible through support provided by the Office of Health and Nutrition, United States Agency for International Development (USAID), under the terms of contract No. HRN-C-00-9300038-00 and John Snow Incorporated (JSI). The contents and opinions expressed herein are those of the authors and do not necessarily reflect the views of USAID and JSI.

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Perceptions, beliefs and practices of mothers in sub-urban and rural areas towards measles and measles vaccination in northern Nigeria

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TROPICAL DOCTOR, 2001, **31**, 89–90

SUMMARY Measles is of particular concern in Nigeria because of the high fatality rate, and high morbidity rate, particularly in young children. Measles and its complications are a common reason for hospitalization, indicating very low immunization coverage. This study was carried out to elucidate the contributing factors from attitudes, beliefs and practices of mothers towards measles and its vaccination. A cross-sectional survey was conducted in Konduga Local Government Area. One per cent of the 500 mothers interviewed believed that measles is prevented by immunization, 16% that it is contagious or due to an infectious agent, 26% that it is caused by evil spirits, witchcraft and heat, and 25% had never heard of measles immunization. Twenty-seven per cent said they did not believe immunization was effective and 4% were not allowed to go for immunization by their husbands. Of those mothers whose children had developed measles, only 31% had been treated in formal health facilities. These results indicate an unfavourable attitude and practice by mothers in relation to measles and measles vaccination. There is the need for an intensive health education campaign to improve this state of affairs and to reduce the morbidity and mortality from measles.

Introduction

Measles is an important cause of morbidity in childhood in many African countries[1–4]. The severity of measles infection is related to the young age at which it is usually contracted and the high prevalence of malnutrition in the developing world[5]. Other factors include lack of immunization, delay in seeking medical treatment[3,4] and the relatively low socio-economic attainments of the population. The complications of measles include developmental retardation, life-long handicap, and both direct and indirect economic loss[6]. The case fatality rate has remained persistently high despite the introduction of EPI in Nigeria (personal communication). The coverage of EPI is now less than the 50% estimated in 1994. The incidence of measles and its complications is increasing, probably because of inadequate implementation of the current measles elimination strategy[7,8]. There has been no community survey as to the perceptions, beliefs and practices of the mothers in relation to measles infection and its vaccination. However, a hospital based study[9] conducted 10 years prior to this study, showed that the low immunization coverage might have resulted from adverse perceptions and beliefs. The long-term aim is to provide information for health education towards improved care-seeking behaviour in order to reduce the morbidity due to measles. In this paper we report a community survey.

Study populations and methods

The study was undertaken in Konduga Local Government Area (LGA) in Borno State, Northeastern Nigeria, population 173,856 (1991 census). The LGA comprises two districts, Konduga and Auno. Kanuris constitute 80% of the population, with the remainder comprising Mafa, Hausa, Fulani and Marghi ethnic groups. The LGA is basically rural, has a general hospital, about 13 dispensaries and five maternity and child welfare centres. The villages have access to the dispensaries, child welfare and maternity centres, especially now that emphasis is on primary healthcare.

Methods

The survey was cross-sectional, conducted over an 8-week period during April and May. Districts in the LGA were divided into clusters of wards. The sample size for this study was 500, to whom 400 questionnaires were administered in Konduga and 100 in Auno districts. Respondents were selected by systematic random sampling. Of the 25 villages, eight villages (Auno, Dalwa-West, Jakana, Wanori, Konduga, Malari, Kawuri, and Mandarari) four from each district, were selected because of population size; they are more densely populated and are representative of most villages in the northern region.

Questionnaires were administered in alternate houses to all potential respondents. The majority of families are polygamous. The choice of alternate houses minimized the possibility of omission of a large number of potential respondents. The cooperation of the communities was enlisted through discussion with the Chairman, traditional rulers and primary healthcare personnel in the LGA; facilitated by the University-community partnership programme between the Department of Community Medicine, University of Maiduguri, and the LGA. Data were collected, based on a preset proforma used by final year medical students during their rural posting in community medicine. The students were supervised by academic and consultant staff in the Department of Paediatrics and Community Medicine, University of Maiduguri, including the authors. The medical students were assisted by trained female interviewers from the primary healthcare department of the LGA where students were not fluent in the language of the respondent. Female interviewers also facilitated access to households, especially where the respondents were Moslems whose husbands would normally restrict interaction with strangers.